

	0	$30^\circ\left(\frac{\pi}{6}\right)$	$45^\circ\left(\frac{\pi}{4}\right)$	$60^\circ\left(\frac{\pi}{3}\right)$	$90^\circ\left(\frac{\pi}{2}\right)$
$\sin$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tg	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$-$
cotg	$-$	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0

$$\sin^2 x + \cos^2 x = 1$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

$$\operatorname{tg}(x) \cdot \operatorname{cotg}(x) = 1$$

$$\operatorname{tg} x = \frac{\sin x}{\cos x}$$

$$\operatorname{cotg} x = \frac{\cos x}{\sin x}$$

$$\sin(-x) = -\sin(x)$$

$$\cos(-x) = \cos(x)$$

D.C.

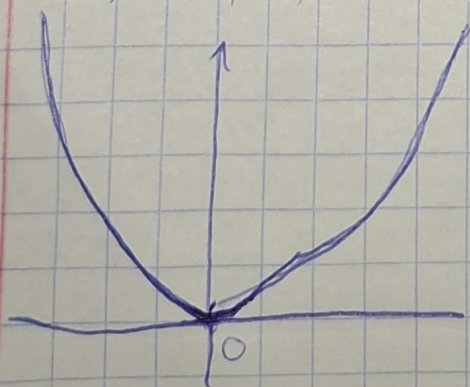
- $\frac{1}{x}$, $x \neq 0$
- $\sqrt{x}$, $x \geq 0$
- $\log x$, $x > 0$

Графики

$$y = x^2$$

$$\bullet x \in (-\infty, +\infty)$$

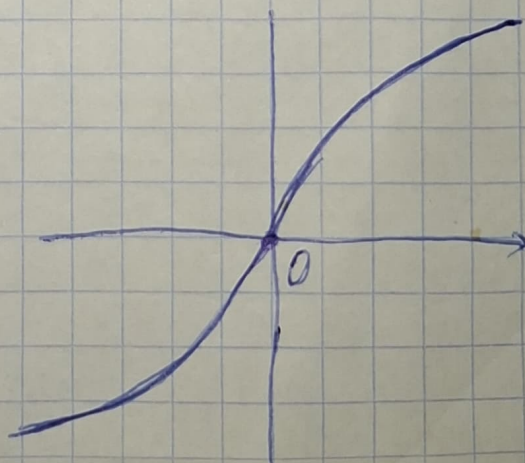
$$\bullet y \in [0, +\infty)$$



$$y = \sqrt[3]{x} = x^{\frac{1}{3}}$$

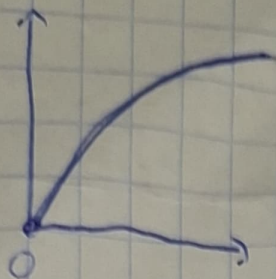
$$\bullet x \in (-\infty, +\infty)$$

$$\bullet y \in (-\infty, +\infty)$$



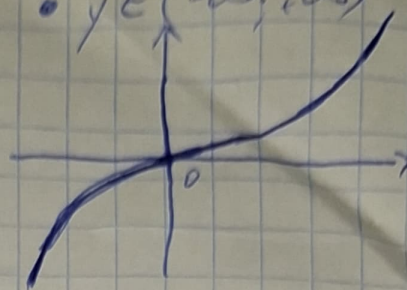
$$y = \sqrt[2]{x} = x^{\frac{1}{2}}$$

- $x \in [0, +\infty)$
- $y \in [0, +\infty)$



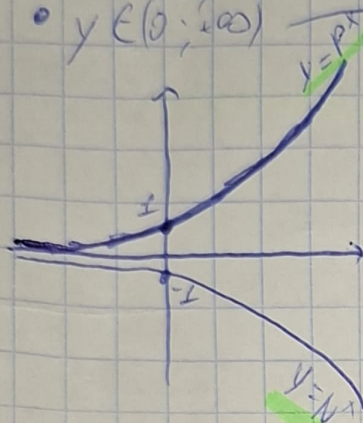
$$y = \sqrt[3]{x} = x^{\frac{1}{3}}$$

- $x \in (-\infty, +\infty)$
- $y \in (-\infty, +\infty)$



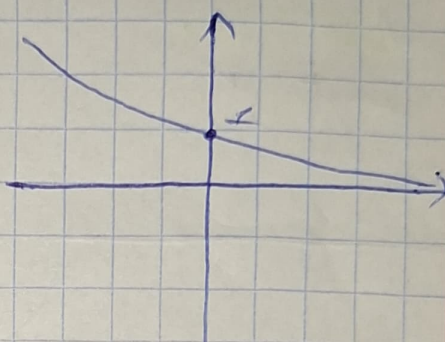
$$y = p^x \quad p = 1, 2, \dots, +\infty$$

- $x \in (-\infty, +\infty)$
- $y \in (0, +\infty)$



$$y = \left(\frac{1}{2}\right)^x$$

- $x \in (-\infty, +\infty)$
- $y \in (0, +\infty)$



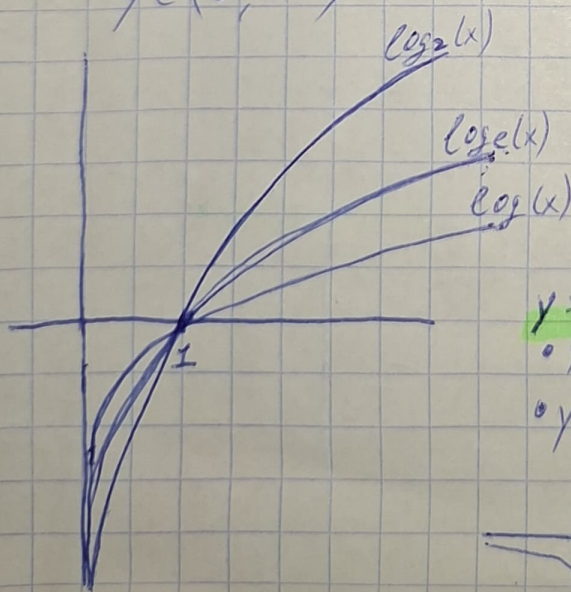
$$y = N^x \quad N = -1, -2, \dots, -\infty$$

- $x \in (-\infty, +\infty)$
- $y \in (0, +\infty)$

$$y = \left(-\frac{1}{2}\right)^x$$

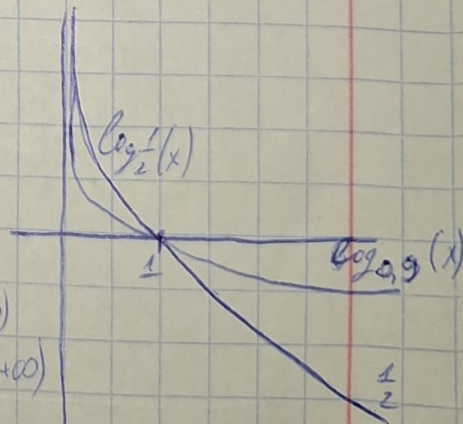
$$y = \log_a x$$

- $x \in (0, +\infty)$
- $y \in (-\infty, +\infty)$



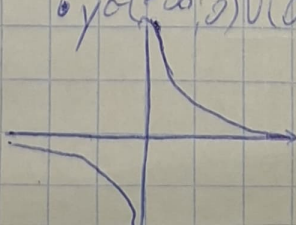
$$y = \log_{\frac{1}{2}} x$$

- $x \in (0, +\infty)$
- $y \in (-\infty, +\infty)$



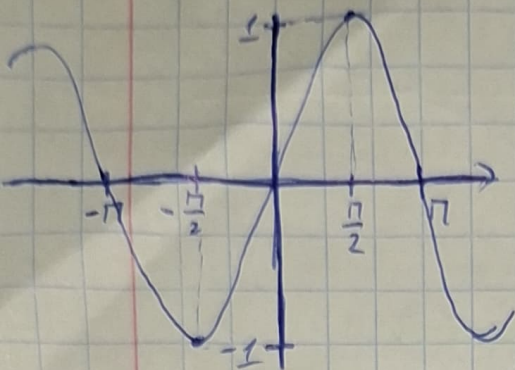
$$y = \frac{1}{x}$$

- $x \in (-\infty, 0) \cup (0, +\infty)$
- $y \in (-\infty, 0) \cup (0, +\infty)$



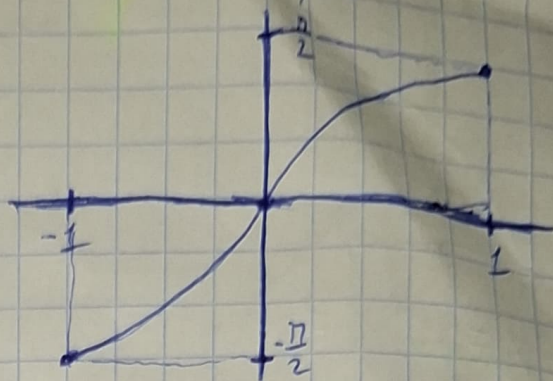
$$y = \sin(x)$$

- $x \in (-\infty, +\infty)$
- $y \in [-1; 1]$



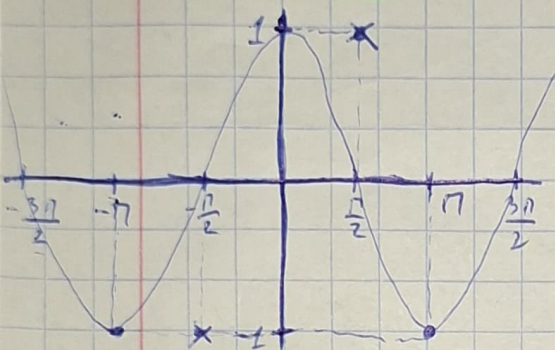
$$y = \arcsin(x)$$

- $x \in [-1; 1]$
- $y \in [-\frac{\pi}{2}; \frac{\pi}{2}]$



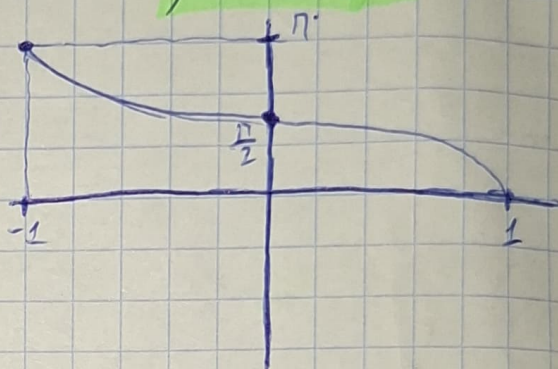
$$y = \cos(x)$$

- $x \in (-\infty; +\infty)$
- $y \in [-1; 1]$



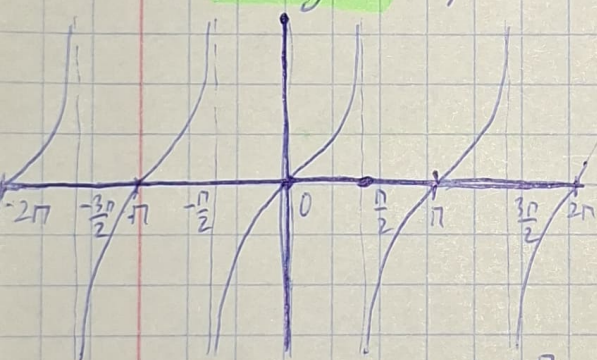
$$y = \arccos(x)$$

- $x \in [-1; 1]$
- $y \in [0; \pi]$



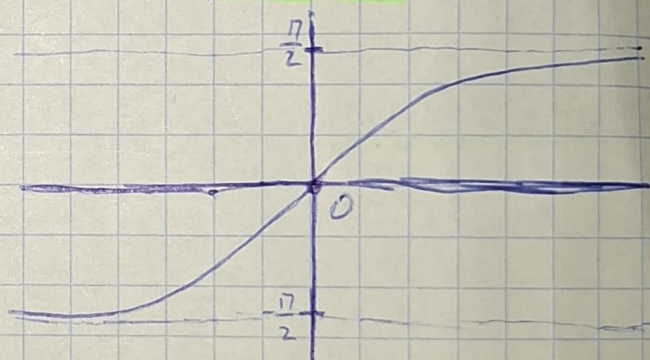
$$y = \tan(x)$$

- $x \in (2k\pi - \frac{\pi}{2}; 2k\pi + \frac{\pi}{2})$
- $y \in (-\infty; +\infty)$



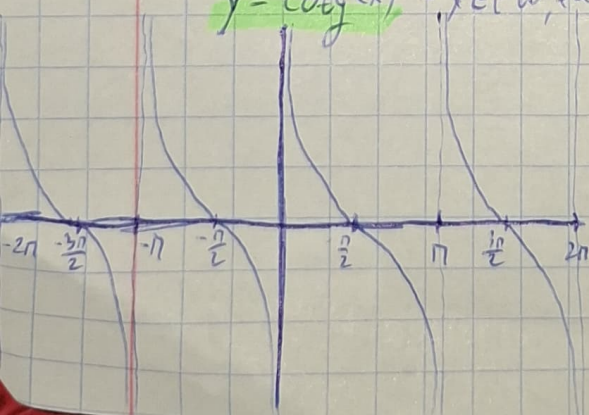
$$y = \arctan(x)$$

- $x \in (-\infty; +\infty)$
- $y \in (-\frac{\pi}{2}; \frac{\pi}{2})$



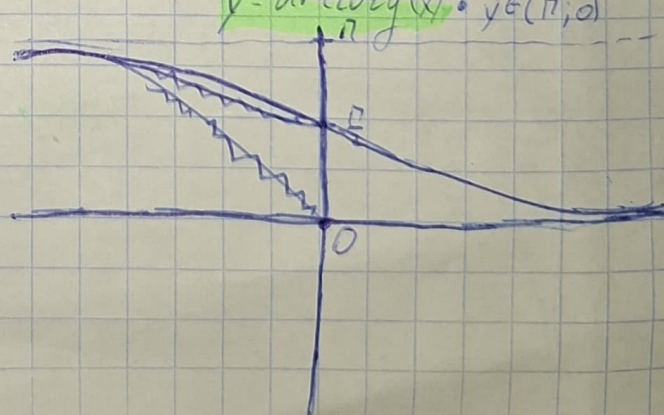
$$y = \cotg(x)$$

- $x \in k\pi$
- $x \in (-\infty; +\infty)$



$$y = \operatorname{arccotg}(x)$$

- $x \in (-\infty; +\infty)$
- $y \in (0; \pi)$



Производные

$f(x)$

- ✓ C
- ✓ x
- ✓ x^n
- ✓ $\sqrt{x} = x^{\frac{1}{2}}$

✓ $\frac{1}{x} = x^{-1}$

✓ e^{ax}

□ a^x

✓ $\ln(x)$

□ $\log_a x$

$f'(x)$

0

1

$n \cdot x^{n-1}$

$\frac{1}{2\sqrt{x}}$

$-\frac{1}{x^2}$

e^x

$a^x \cdot \ln(a)$

$\frac{1}{x}$

x

$\frac{1}{x \cdot \ln(a)}$

$\cos(x)$

$-\sin(x)$

1

$\cos^2 x$

1

$\sin^2 x$

1

$\sqrt{1-x^2}$

1

$\sqrt{1-x^2}$

1

$1+x^2$

1

$1+x^2$

пр

$5^1 = 0$

$x^1 = 1$

$(x^4)' = 4x^3$

$(\sqrt{4x})' = \frac{1}{2\sqrt{x}} \cdot (4x)'$

$(\frac{4}{4x})' = -\frac{4}{(4x)^2} \cdot (4x)'$

$(e^x)' = e^x$

$(2^x)' = 2^x \cdot \ln(2)$

$(\ln(4x))' = \frac{1}{4x} \cdot (4x)' = \frac{1}{x}$

$(\log_2 x)' = \frac{1}{x \cdot \ln 2}$

$x \rightarrow 0$

$\sin(x) \sim x$

$\tan(x) \sim x$

$\arcsin(x) \sim x$

$\arctg(x) \sim x$

$e^x - 1 \sim x$

$\ln(1+x) \sim x$

$\ln t \sim t-1, t \rightarrow 1$

$f(3x) \sim 3x$

$e^{3x} \sim$

Правила

$(C \cdot f)' = C \cdot f'$

$(f \pm g)' = f' \pm g'$

$(f \cdot g)' = f'g + fg'$

$(\frac{f}{g})' = \frac{f'g - fg'}{g^2}$

$f(g(x))' = f'(g(x)) \cdot g'(x)$

$(5x^3)' = 5 \cdot 3x^2 = 15x^2$

$(x^2 + \sin x)' = 2x + \cos x$

$(x^2 \cdot e^x)' = 2x e^x + x^2 e^x$

$\frac{\sin x}{x} = \frac{\cos x \cdot x - \sin x \cdot 1}{x^2}$

$$-\sqrt{3} \quad -1 \quad -\frac{\sqrt{3}}{2} \quad -\frac{\sqrt{2}}{2} \quad -\frac{\sqrt{3}}{3} \quad -\frac{1}{2} \quad 0 \quad \frac{1}{2} \quad \frac{\sqrt{3}}{3} \quad \frac{\sqrt{2}}{2} \quad \frac{\sqrt{3}}{2} \quad 1 \quad \sqrt{3}$$

$$\arcsin = -\frac{\pi}{2} \quad -\frac{\pi}{3} \quad -\frac{\pi}{4} \quad - \quad -\frac{\pi}{6} \quad 0 \quad \frac{\pi}{6} \quad - \quad \frac{\pi}{4} \quad \frac{\pi}{3} \quad \frac{\pi}{2} \quad -$$

$$\arccos = \pi \quad \frac{5\pi}{6} \quad \frac{3\pi}{4} \quad - \quad \frac{2\pi}{3} \quad \frac{\pi}{2} \quad \frac{\pi}{3} \quad - \quad \frac{\pi}{4} \quad \frac{\pi}{6} \quad 0 \quad -$$

$$\operatorname{arctg} -\frac{\pi}{3} \quad -\frac{\pi}{4} \quad - \quad - \quad -\frac{\pi}{6} \quad - \quad 0 \quad - \quad \frac{\pi}{6} \quad \frac{\pi}{4} \quad - \quad \frac{\pi}{3}$$

$$\operatorname{arccotg} \frac{5\pi}{6} \quad \frac{3\pi}{4} \quad - \quad - \quad \frac{2\pi}{3} \quad - \quad \frac{\pi}{2} \quad - \quad \frac{\pi}{3} \quad - \quad - \quad \frac{\pi}{4} \quad \frac{\pi}{6}$$